

EXPERIMENT-2

1. Print this Pattern:

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InvertedPyramidPattern.java	Run	Output
<pre>1 public class InvertedPyramidPattern { 2 public static void main(String[] args) { 3 int n = 5; 4 5 for (int i = 0; i < n; i++) { 6 for (int j = 0; j < 2 * i; j++) { 7 System.out.print(" "); 8 } 9 for (int j = 0; j < n - i; j++) { 10 System.out.print("*"); 11 } 12 System.out.println(); 13 } 14 } 15 }</pre>		<pre>***** **** *** ** *</pre>

2. Write a Java program that calculates the electricity bill based on the number of units consumed by a user. The electricity charges are determined as follows:

For consumption between 1 and 100 units, the cost is Rs 5 per unit.

For consumption between 101 and 200 units, the cost is Rs 7 per unit.

For consumption between 201 and 300 units, the cost is Rs 10 per unit.

For consumption exceeding 300 units, the cost is Rs 15 per unit.

The program should:

Take the total units consumed as input.

Calculate the total electricity bill based on the charges above.

Display the total electricity bill.

Example Input/Output:

Input: 150

Output: The total electricity bill is Rs 850.

ElectricityBillCalculator.java	Output
<pre>1 public class ElectricityBillCalculator { 2 public static void main(String[] args) { 3 System.out.print("Enter total units consumed: "); 4 int units = Integer.parseInt(System.console().readLine()); 5 6 int bill = 0; 7 8 if (units <= 100) { 9 bill = units * 5; 10 } else if (units <= 200) { 11 bill = (100 * 5) + ((units - 100) * 7); 12 } else if (units <= 300) { 13 bill = (100 * 5) + (100 * 7) + ((units - 200) * 10); 14 } else { 15 bill = (100 * 5) + (100 * 7) + (100 * 10) + ((units - 300) 16 * 15); 17 } 18 System.out.println("Total electricity bill: Rs " + bill); 19 } 20 }</pre>	<pre>Enter total units consumed: 150 Total electricity bill: Rs 850 === Code Execution Successful ===</pre>

3. Write a program to find the area of a shape (Rectangle, Square, Circle, Triangle) using method overloading.

ShapeAreaCalculator.java	Output
<pre>1 public class ShapeAreaCalculator { 2 3 public static double area(double length, double breadth) { 4 return length * breadth; 5 } 6 7 public static double area(double side) { 8 return side * side; 9 } 10 11 public static double areaCircle(double radius) { 12 return Math.PI * radius * radius; 13 } 14 15 public static double areaTriangle(double base, double height) { 16 return 0.5 * base * height; 17 } 18 19 public static void main(String[] args) { 20 System.out.println("Area of Rectangle: " + area(10, 5)); 21 System.out.println("Area of Square: " + area(4)); 22 System.out.println("Area of Circle: " + areaCircle(7)); 23 System.out.println("Area of Triangle: " + areaTriangle(6, 8 24)); 25 } 26 }</pre>	<pre>Area of Rectangle: 50.0 Area of Square: 16.0 Area of Circle: 153.93804002589985 Area of Triangle: 24.0 === Code Execution Successful ===</pre>

4. Write a Java program that takes an operator (+, -, *, /, %) and two numbers as input from the user. Use a switch statement to perform the specified operation and display the result.

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Calculator.java                                                                                                                                                                                                                                                                                                                                                                                                                                   
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